Discrete Mathematics

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MODULE 5
BASICS OF COUNTING

Introduction

The problem of counting the number of different objects appears in many computer science applications. Some examples are all Java programs use identifiers. Say, each identifier is limited to 8 characters long, the first character must be a letter. How many possible identifiers are there? In GPS map routing application, we know the starting location, the ending location, and the intermediate locations on the map, how many different routes are there? The routing program must select the best route from these routes. In Facebook application, we are given three users A, B, C. We know the set of the friends of A, B, C. How do we count the number of users who are the friend of at least one of A, B, C?

According to Wheelan, C. (2014), the fundamental counting principle which is called the counting rule is a way to figure out the number of outcomes in a probability problem. Basically, you multiply the events together to get the total number of outcomes. The formula that can be considered is: *If you have an event "a" and another event "b" then all the different outcomes for the events is a * b.* In this module we will discuss some commonly used counting methods.

Learning Outcomes

At the end of this module, students should be able to:

1. Describe the principle of counting;
2. Analyze examples of principle of counting; and
3. Apply counting principles to determine probabilities.

Lesson 1. Fundamental Counting Principle

If there are m ways to do one thing, and n ways to do another, then there are m*n ways of doing both (Gonzales, A., n.d).

The Fundamental Counting Principle is the guiding rule for finding the number of ways to accomplish two tasks.

Examples using the counting principle:

Let's say that you want to flip a coin and roll a die. There are 2 ways that you can flip a coin and 6 ways that you can roll a die. There are then 2x6=12 ways that you can flip a coin *and* roll a die.

If you want to hit one note on a piano and play one string on a banjo, then there are 88 * 5 = 440 ways to do both.

If you want to draw 2 cards from a standard deck of 52 cards without replacing them, then there are 52 ways to draw the first and 51 ways to draw the second, so there are a total of 52*51 = 2652 ways to draw the two cards.

Sample Spaces

A listing of all the possible outcomes is called the sample space and is denoted by the capital letter S.

The sample space for the experiments of flipping a coin and rolling a die are S = { H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}. Sure enough, there are twelve possible ways. The fundamental counting principle allows us to figure out that there are twelve ways without having to list them all out.

Lesson 2: Counting Arguments

As mentioned by Levy (n.d) in his article, people often use counting arguments, which are based on deducing things from sizes of sets.

For example, suppose we have a function from a set with *n* elements to another set with *n* elements. This function is 1-to-1 iff it's onto (1-to-1, or injective, means that $f(x) != f(y)$ when $x != y$; onto, or surjective, means that every element in the range can be written as $f(x)$ for some x; invertible, or bijective, means 1-to-1 and onto).

To see why, let's say that the function *f* from X to Y is not onto. There is some *y* in Y that isn't of the form *f*(*x*) for some *x* in X. So the image of *f* has at most *n*-1 elements. But X has *n* elements, so the function can't be 1-to-1. This is called the pigeonhole principle: if there are more pigeons than pigeonholes, at least two pigeons must share a pigeonhole. Similarly, if the function is not 1-to-1, we must have *f*(*x*1) = *f*(*x*2) for some distinct *x*1 and *x*2 in X. But then the image has at most *n*-1 elements, whereas Y has *n*.

In algebra, it's relevant because it implies that in a finite ring, every element that isn't a zero divisor is a unit. If *a* satisfies the condition that *ab* = 0 implies *a* = 0, then the function from R to itself that sends *x* to *ax* is 1-to-1. This is because *ax* = *ay* implies *a*(*x*–*y*) = 0, so *x*–*y* = 0, i.e. *x* = *y*. Then the function is onto, and we can find some *b* with *ab* = 1. In particular, if R is a finite integral domain, it's a field. I should add that this requires no additional assumptions; the existence of *a* implies that R has 1, and even if R is not commutative, the condition implies that *ab* = *ba* = 1 for some *b*.

The Rules of Sum and Product

The **Rule of Sum** and **Rule of Product** are used to decompose difficult counting problems into simple problems.

- **The Rule of Sum** – If a sequence of tasks $T_1, T_2, \ldots, T_m$ can be done in $w_1, w_2, \ldots w_m$ ways respectively (the condition is that no tasks can be performed simultaneously), then the number of ways to do one of these tasks is $w_1 + w_2 + \cdots + w_m$. If we consider two tasks A and B which are disjoint (i.e. $A \cap B = \emptyset$), then mathematically $|A \cup B| = |A| + |B|$

Sum Rule

Let $A_1, A_2, \ldots, A_m$ be different sets such that $A_i \cap A_j = \emptyset$ for all $i \neq j$. Then:

$$|A_1 \cup A_2 \cup \ldots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m|$$

Figure 5.1 Sum Rule

Source: https://cse.buffalo.edu/

Example:

In a store, there are 56 salted eggs, 105 unsalted eggs. Assuming customer buys both salted and unsalted eggs, how many customers bought either salted or unsalted eggs?

Solution: Simple enough: 56+105= 161. It is simple because we assumed Ai ∩ Aj = ∅

Inclusion-Exclusion Principle (with two sets)

Inclusion-Exclusion Principle (with two sets)

Let A_1, A_2 be two sets. Then:

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$

Figure 5.2 Inclusion-Exclusion Principle

Source: https://cse.buffalo.edu/

Example:

In a store, customers bought 56 salted eggs, 105 unsalted eggs. 25 customers bought both salted and unsalted eggs, how many customers bought either salted or unsalted eggs?

Solution: Simple enough: 56+105-25 = 136

Note: This principle can be generalized into two or more sets.

- **The Rule of Product** – If a sequence of tasks $T_1, T_2, \ldots, T_m$ can be done in $w_1, w_2, \ldots w_m$ ways respectively and every task arrives after the occurrence of the previous task, then there are $w_1 \times w_2 \times \cdots \times w_m$ ways to perform the tasks. Mathematically, if a task B arrives after a task A, then $|A \times B| = |A| \times |B|$

Example:

Question – A boy lives at X and wants to go to School at Z. From his home X he has to first reach Y and then Y to Z. He may go X to Y by either 3 bus routes or 2 train routes. From there, he can either choose 4 bus routes or 5 train routes to reach Z. How many ways are there to go from X to Z?

Solution − From X to Y, he can go in 3+2=53+2=5 ways (Rule of Sum). Thereafter, he can go Y to Z in 4+5=94+5=9 ways (Rule of Sum). Hence from X to Z he can go in 5×9=455×9=45 ways (Rule of Product).

The product rule can be easily generalized from 2 tasks to k tasks for any integer k.

Example:

How many different bit strings of length 8 are there?

Solution $2^8 = 256$

Assessment Task

Direction: Kindly answer all the assessment task and submit it on/or before the midterm examination through your account in Edmodo. If hand written, write it legibly and clearly, take a picture of it and send it in a MS Word format. Use the sample filename – CS2108AT_JABanocnoc for BSCS Students and IT2109AT_JABanocnoc for BSIT students:

1. The chairs in an auditorium are labeled by upper case English letters followed by an integer between 0 and 99. What is the largest numbers of chairs that can be labeled this way?
2. Suppose that Java program identifiers consist of only upper and lower cases English letters and digits. The first character must be an uppercase letter. The last character must be a digit. Count the number of possible identifiers of length exactly 6.
3. In a high school, 32 students take French class, 45 students take Spanish. Assuming no student take both French and Spanish, how many students take either French or Spanish?
4. In a high school; 32 students take French class; 45 students take Spanish. 15 students take both French and Spanish. How many students take either French or Spanish?
5. In a high school, 32 students take French class, 45 students take Spanish, 20 take German. 15 students take both French and Spanish; 12 students take both German and Spanish; 10 students take both French and German; 6 students take all three foreign languages. How many students take at least one foreign language?

Summary

In this module, we have learned the concepts of counting in discrete mathematics. Different ideas were explained to understand we can determine probabilities. We also learned the principles of pigeonhole which can help us to tackle certain combinatorial problems in a manner that is more elegant and systematic.

This study of basic counting numbers is the root of several topics in mathematics such as number theory probability, discrete mathematics, graph theory, and many other specialties.

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MODULE 6
PIGEONHOLE PRINCIPLE

Introduction

In the world of computing, there are so many methodologies, laws and principles that a computer scientist needs to analyze and follow to be able to derived at a best output. In this module we will take a peek of the lessons associated with Pigeonhole principle. Most of the students in the lower year still do not understand the use of it and do not know at least right now where to apply this. Some maybe wondering, where would the pigeon hole principle be useful in Computer Science. Does it really take a student to exert effort in studying this type of principle? The idea of Pigeonhole Principle is very simple. But when used properly, we can get very interesting results.

The first formalization of the pigeonhole concept is believed to have been made by Dirichlet (1805–1859) as what he called Schubfachprinzip or the “drawer/shelf principle”. As Dirichlet published works in both German and French, he would alternate between calling the principle Schubfach and tiroir, which both translate to drawer. However, as Dirichlet’s father was a postmaster it is believed that the type of drawer he was referring to might have best been translated to English as pigeon-hole, such as those commonly used for storing and sorting mail. The first appearance of the term “pigeonhole principle” was used by mathematician Raphael M. Robinson in 1940 (Veisdal, 2019).

Learning Outcomes

At the end of this module, students should be able to:

1. Identify the Pigeonhole Principle;
2. Explain the relationship of Pigeonhole principle to finite set and functions;
3. Compute examples of pigeonhole principle; and
4. Examine the pigeonhole proofs.

Lesson 1. Pigeonhole Principle Defined

The pigeonhole principle is a simple, yet beautiful and useful idea. Given a set A of pigeons and a set B of pigeonholes, if all the pigeons fly into a pigeonhole and there are more pigeons than holes, then one of the pigeonholes has to contain more than one pigeon. The pigeonhole principle states that if n items are put into m containers, with n > m, then at least one container must contain more than one item (Veisdal, 2019).

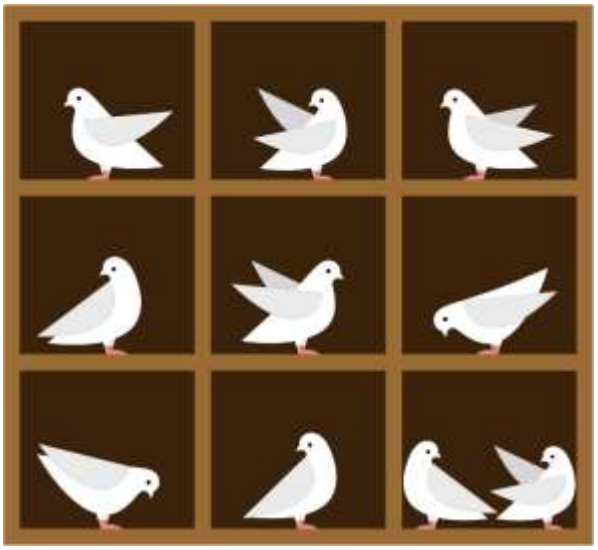


Figure 6.1 Simple illustration of Pigeonhole Principle

Source: https://medium.com/cantors-paradise/the-pigeonhole-principle-e4c637940619

Georg Cantor (1845–1918) indirectly made use of the pigeonhole principle in his 1873–74 investigations of countability (Veisdal, 2019):

For practical reasons let us illustrate the finite case by imagining a classroom with 100 seats. Filled with students, one can make an inference about the size of the set of students in relation to size of the set of seats. If seats are vacant, the set of seats is larger than the set of students. If no seats are vacant and some students are standing, the size of the set of students is larger than that of seats, and so on.- Excerpt, "The Nature of Infinity and Beyond" by Veisdal (2018)

Veisdal (2019) said that although a very simple and easily clear concept, the pigeonhole principle is powerful enough to prove truths that are far from obvious. In a sense one of the quintessential counting arguments, the pigeonhole principle for instance allows us to prove statements such as:

- At any given time there live at least two people in California with the same number of hairs on their heads;
- You cannot cover an 8 x 8 square chess board with two diagonally opposite corners removed, with domino pieces of size 1 x 2;
- In a room full of 367 people there are at least two people with the same birthday;
- Any lossless compression algorithm that makes some inputs smaller will also make some outputs larger;

Finite Set

If you have a function from a finite set to a smaller finite set, then the function cannot be one-to-one; in other words, there must be at least two elements in the domain with the same image in the codomain.

If $f{:}X \rightarrow Y$ where X and Y are finite sets with $|X|>|Y|$, then f is not one-to-one

Definition

Formally, think of the pigeonhole principle as the statement about a function f from domain P → PH, where pigeon n flies into pigeonhole f(n), as is shown in the figure below:

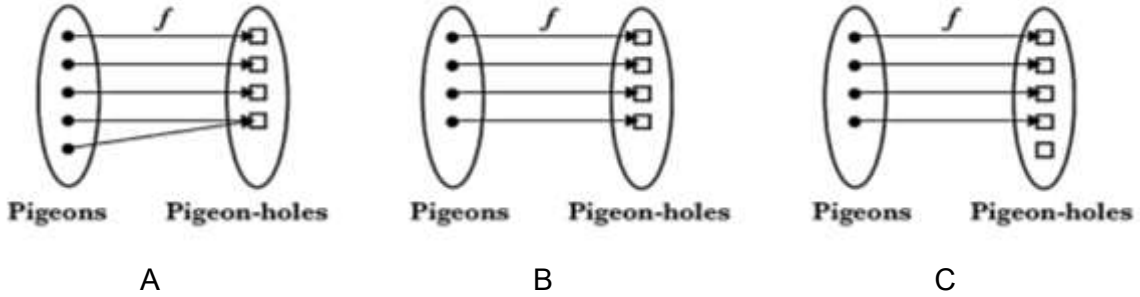


Figure 6.2 Functions from a Domain to a Codomain

Source: https://medium.com/cantors-paradise/the-pigeonhole-principle-e4c637940619

The three cases illustrate three types of functions from a domain ("pigeons") to a codomain "pigeonholes" which can be mapped to one another:

Case A: If |n| > |f(n)|, then the function is a non-injective surjective function (not a bijection)
There are more pigeons than the pigeonholes |n| > |f(n)|

Case B: If |n|= |f(n)|, then the function is an injective surjective function (bijection)
Same number of pigeons as pigeonholes |n| = |f(n)|

Case C: If |n| < |f(n)|, then the function is an injective non-surjective function (not a bijection).
More pigeonholes than pigeons |n| < |f(n)|

Case A is the case illustrating the core of the pigeonhole principle, namely that if you put n items into m containers and n > m, then at least one container must contain more than one item. The function f that maps pigeons into pigeonholes is hence called a surjective function. We may prove the principle easily by contradiction:

Proof of Pigeonhole Principle

Suppose that the total number of pigeons n are to be put in m number of pigeonholes and n > m. Assume that there are no pigeonholes with at least n/m pigeons, i.e. that each pigeonhole has less than n/m pigeons:

Remember: n = pigeons and m = pigeonholes

Number of pigeons in each pigeonhole < n/m

Number of pigeons < m × n/m

Number of pigeons < n

But given that the number of pigeons is strictly equal to n, we have obtained a contradiction to our assumption that each pigeonhole has less than n/m pigeons. Hence there exists at least one pigeonhole with at least n/m pigeons.

Pigeonhole Principle Examples:

Generalized Pigeonhole Principle

If you put n pigeons into m pigeonholes, then at least one pigeonhole contains at least $\lceil n/m \rceil$ pigeons.

Let: $\lceil x \rceil$ denotes the smallest integer that is ≥ x. Here $\lceil x \rceil$ is the **ceiling function**, which represents the round-up values of x

Sample Answers:

[0.6] = 1

[3] = 3

[4.0000001] = 5

Other Examples:

Example 1:

At least 2 people in the Philippines have the same number of hair.

Example 2:

At least 16 people in the Philippines have the same number of hair.

Answer:

- There are 8 Million people in the Philippines
- It is known that the maximum number of hair a person can have is 500,000
- Each person is a "pigeon"
- The "ith pigeonhole" hold the people with i hair. So there are 500,001 "pigeonholes"
- So at least 1 pigeonhole contains at least [8,000,000/500,001] =[15.999968] = 16 people

Example 3:

Let's say as of today, at least 1574 cellphones are connected to PLDT cellphone tower in the Philippines

- There are 40,032 PLDT towers, and 63,000,000 cellphones (note: the numbers are not facts they are only an example)
- So at least [63,000,000/40,032] = 1574 cellphones are connected to the tower

More difficult examples:

1. At least two people in this room have the same number of friends in this room. (Of course, we must assume there are at least two people in this room.)

   Proof: Let k be the number of people who has at least one friend in this room.
   Case 1 $k = 0$: Then all people have 0 friends.
   Case 2 $k > 0$: Among these k people:
   - the numbers of friends they can have are $1, 2, \ldots, (k-1)$. These choices are $k-1$ pigeonholes.
   - But there are k people, each is a "pigeon".
   - So at least one pigeonhole contains at least two pigeons. In other words, at least two people have the same number of friends.

2. $\pi \approx 3/1 = 3$
   $\approx 22/7 = 3.1428571\ldots$
   $\approx 223/71 = 3.1408450\ldots$
   $\approx 333/106 = 3.1415094\ldots\ldots$

   Proof: For any number x, let {x} denote the fractional part of x, (i.e. the part after the decimal point.)
   Example:
   - $\{0.34\} = 0.34$.
   - $\{100.28\} = 0.28$.
   - $\{\pi\} = 0.1415926\ldots$.

   [c, d] denotes the interval of real numbers x such that $c \le x < d$

Lesson 2. Application

As discussed in the previous lesson although simple to understand, the pigeonhole principle allows us to prove statements that sometimes seem unknowable. Statements such as:

Example 1: Common Properties in Large Groups

At least 2 people in the Philippines have the same number of hairs.

To prove this, we need to establish two facts. First, that the population of the Philippines is larger than 20 million (actually it's more like 40 million). Second, that it is a biological fact that every human head has fewer than one million hairs (typically, the number ranges between 90,000–150,000):

Proof of Example 1 (Hammack, 2013 pp. 207)

Let A be the set of all Filipino heads and B = {0,1,2,3,4, ..., 1000000} be the number of hairs one can have on ones' head. Let f be a function that maps A → B, for which f(x) equals the number of hairs on the head of x. Since |A|>|B|, the pigeonhole principle asserts that f is not injective. Thus, there are two Filipinos x and y for whom f(x) = f(y), meaning they have the same number of hairs on their heads.

Example 2: Subset Cardinalities

If A is any set of 10 integers between 1 and 100, then there must exist two different subsets X and Y for which the sum of elements in X equals the sum of the elements in Y.

Consider for instance the random set A = {6, 7, 11, 12, 17, 50, 51, 80, 90, 100} which has subsets X = {7, 12, 17, 50} and Y = {6, 80} for which the sum of the elements in X equals the sum of those in Y, namely 86. If we try to mess this up by changing a value in A (say, changing the 7 to a 5), we would instead get A' = {5, 6, 11, 12, 17, 50, 51, 80, 90, 100} and yet still find subsets W = {6, 12, 17, 50} and Z = {5, 80} whose cardinalities are both equal to 85.

Proof of Example 2 (Hammack, 2013 pp. 206)

Suppose A ⊆ {1, 2, 3, 4, ..., 99, 100} and |A| = 10.

Notice that if X ⊆ A, the X has no more than 10 elements, each between 1 and 100. Therefore, the sum of all the elements of X is less than 100 × 10 = 1000.

Consider the function f: $\mathcal{P}$(A) → {0,1,2,3,4,...,1000}

Where: f(X) is the sum of the elements in X.

As the cardinality of |$\mathcal{P}$(A)| = 2^{10} = 1024 > 1001 = |{0,1,2,3,...,1000}| it follows from the pigeonhole principle that f is not injective, i.e. that it is a "many-to-one" function. Therefore, there

are two unequal sets X, Y $\in \mathcal{P}$(A) for which f(X) = f(Y). Stated differently, there are subsets X ⊆ A and Y ⊆ A for which the sum of elements in X equals the sum of elements in Y.

Example 3: The Chinese Remainder Theorem (Veisdal, 2019)

The well-known Chinese remainder theorem gives the necessary conditions for multiple equations to have a simultaneous integer solution. It may be stated as:

Let m and n be relatively prime positive integers.
Then the system x = a (mod m), x = b (mod n) has a solution.

To illustrate this practically, think of a collection of balls arranged in a row:

1. If the 52 balls are arranged in rows of 3 there is one ball left over.

   52 ÷ 3 = 17.3333 r. 1

   3 balls per column

   17 balls per row, so there is 1 remaining ball

2. If the 52 balls are arranged in rows of 5 there are 2 balls left over.

   52 ÷ 5 = 10.44 r. 2

   5 balls per column

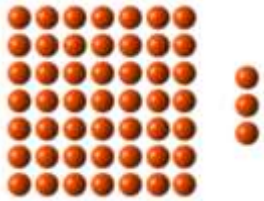
   10 balls per row, so there are 2 remaining balls

3. If the 52 balls are arranged in rows of 5 there are 2 balls left over.

   52 ÷ 7 = 7.428 r. 3

   7 balls per column

   7 balls per row, so there are 3 remaining balls



Conclusion:

The Chinese Remainder Theorem proves that the smallest number of balls must be 52

Veisdal (2019) explained that the Chinese remainder theorem is useful because it establishes that if we know the remainders of the division of an integer n by several integers, then we can determine uniquely the remainder of the division of n by the product of these integers. The theorem was famously first stated as early as the third century by Chinese mathematician Sun-tzu:

Proof of Example C (Veisdal, 2019)

First consider the sequence

(1) $b, b + m, b + 2m, ..., b + (n - 1)m$

The sequence forms a complete residue class, modulo n.

To see this, consider the sequence:

(2) $0, m, 2m, 3m, ..., (n - 1)m$

Since the greatest common divisor of $(m,n) = 1$, we have $xm \equiv ym \pmod{n} \iff x \equiv y \pmod{n}$. Hence, each element in the sequence (2) is unique. Adding b to the sequence (1) only cyclically permutes the remainder class.

Note: The symbol ≡ is read as "congruent of"

By the pigeonhole principle,

for any $0 \leq a < n$, we must have $b + km \equiv a \pmod{n}$ for any $k \in \{0,...,n - 1\}$.

Constructive vs Non-Constructive Proofs

Veisdal (2019) noted about the type of proofs pigeonhole proofs belong to. Although useful, a defining characteristic of pigeonhole proofs are that they are non-constructive, i.e. that although they prove the existence of something, they fail to provide an explicit example that proves the theorem. To illustrate the difference, consider two proofs of the same statement, namely that:

There exist irrational numbers x, y for which x^y is rational

Let us first consider a non-constructive proof (Hammack, 2013 pp. 128) that shows that there exist irrational numbers x and y for which x^y is rational without actually producing an example:

Non-Constructive Proof

Let x = √2^(√2) and y = √2. We know y is irrational, but we do not know whether x is rational or irrational. If x is irrational, then we have an irrational number to an irrational power that is rational:

x^y = (√2^(√2))^(√2) = (√2)^(√2 × √2) = $(\sqrt{2})^2$ = 2

If x is rational, then y^y = √2^(√2) = x is rational.

Either way, we have an irrational number to an irrational power that is rational.

Next consider an alternative, constructive proof:

Constructive Proof

Let x = √2 and y = log(9). Then

x^y = √2^(log(9)) = √2^(log(3^2)) = √2^(2log(3)) = (√2^2)^(log(3))^= 2^(log3) = 3

As 3 is rational, we have shown that x^y = 3 is rational.

We know that x = √2 is irrational. We do not know whether y = log(9) is irrational. Suppose log(9) is rational, so that there are two integers a, b for which a/b = log(9). Then 2^(a/b) = 9, so $2^{(a/b)^b} = 9^b$, which reduces to $2^a = 9^b$. However, we know that 2^a is even and 9^b must be odd, so they cannot equal one another. We have obtained a contradiction and must assume that log(9) is irrational.

In the above proof of the same statement, in the process of constructing the proof we also provide an example of a case where the statement is true, namely for x = √2 and y = log(9). This, one never obtains from pigeonhole- and other non-constructive proofs alone.

Assessment Task

Direction: Kindly answer all the assessment task and submit it on/or before the midterm examination through your account in Edmodo. If hand written, write it legibly and clearly, take a picture of it. If you have a laptop of desktop send it in a MS Word format. Use the sample filename – CS2108AT_JABanocnoc for BSCS Students and IT2109AT_JABanocnoc for BSIT students:

1. Without looking, you pull socks out of a drawer that has just 5 blue socks and 5 white socks. How many do you need to pull to be certain you have two of the same color?
2. in a group of 30 people, each person picks a number from 1 to 10. What is the minimum number of people needed to be sure you have two people with the same number?
3. In a group of 100 people, each person picks a number from 100 to 120. What is the minimum number of people you need to be sure two of them have the same number?
4. How many cards do you need to pick from a standard 52-card deck to be sure to get a red card?
5. How many integers do you need to pick to be sure that at least two of them have the same remainder after dividing by 5?
6. How many students do you need in a school to guarantee that there are at least 2 students who have the same first two initials using the 26 letters A, B, C, ..., Z, ?
7. A school has n=677 students. What is the smallest n such that at least two of the students have matching first and last initials?

Summary

At first glance, the pigeonhole principle (also known as Dirichlet's principle in honor of the eponymous German mathematician) might appear to be too obvious to be useful; indeed, the power of the principle comes from cleverly choosing the "boxes" and "objects." Even though the principle itself is quite simple, it is not always clear if it is useful. The Pigeonhole Principle says if you have more pigeons than pigeonholes, at least 2 pigeons must cuddle up.

Of three ordinary people, two must have the same sex! D. J. Kleitman.

More generally, if n+1 letters are distributed among n letter boxes, then at least one of the letter boxes will have at least two letters. In an abstract manner, if a set consisting of a “large” number of elements, is decomposed into a sufficiently small number of subsets then at least one of the subsets will have many elements of the set.

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MODULE 7
PERMUTATION AND COMBINATIONS

Introduction

Combinatorics is something that shows up everywhere. Permutations and combinations are very useful in computer science. It is prerequisite to Graph theory, Probability etc. Discrete mathematics is one of the most important courses in computer science and permutations and combinations is must to learn. There are so many applications to graph theory and even some uses many concepts of graph theory. Probability and permutations and combinations are very much interlinked and are useful everywhere. More than anything It is also fun to learn and use permutations and combinations (Kakarala, 2014).

As Kakarala (2014) has pointed out, it is one of the many foundations of Discrete Mathematics which is really important in Computer Science. So, what is the practical real-life use of permutation and combination? On the following lessons we will take a quick tour on how permutation and combinations are done. How it is used in our everyday lives particularly in information technology and computer science.

Learning Outcomes

At the end of this module, students should be able to:

1. Identify Permutation and Combination;
2. Explain the relationship of permutation to combination;
3. Compute examples of permutation and combination; and
4. Examine the permutation and combination theorems.

Lesson 1. Permutation

First, anyone reading this should know that a **permutation** refers to an arrangement (i.e. order matters) of objects (Jain, 2017).

- An ordered arrangement of r elements of a set is called an r -permutation
- The number of r-permutations of a set with n elements is denoted by $P(n,r)$

Example 1: List all permutations of the letters ABCD

ABCD	BACD	CABD	DABC
ABDC	BADC	CADB	DACB
ACBD	BCAD	CBAD	DBAC
ACDB	BCDA	CBDA	DBCA
ADBC	BDAC	CDAB	DCAB
ADCB	BDCA	CDBA	DCBA

Example 2:

In how many ways can we select **3 students** from a group of **5 students to stand in line** for a picture?

Solution:

- We can select any of the **5 students** for the first position.
- After the first position is selected, we can choose any of the **4 remaining students** for the 2nd position.
- After the 1st and the 2nd positions are selected, we can choose any of the **3 remaining students** for the 3rd position.

  So, the answer is: $5 \cdot 4 \cdot 3 = 60$. In this example, we are counting the number of distinct **3-permutations** in a set of **5 elements**. What we have shown is:

$$P(5, 3) = 5 \cdot 4 \cdot 3 = 60$$

3 students

Group of 5 students to stand in line

Example 3:

How many ways are there to select a first-prize winner; a second-prize winner; and a third-prize winner from 100 different people?

Solution: $\boldsymbol{P}(100, 3) = 100 \cdot 99 \cdot 98 = 970200$.

Theorem 1

The factorial of n is denoted by n! and defined by:

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \cdot \cdot 2 \cdot 1$$

Note: For convenience, we define: 0!=1.

Theorem 1:

$$\begin{aligned} P(n,r) &= \underbrace{n \cdot (n-1) \cdots (n-r+1)}_{r \text{ terms}} = \frac{n!}{(n-r)!} \\ P(n,n) &= n \cdot (n-1) \cdots 2 \cdot 1 = n! \end{aligned}$$

Figure 7.1 Formula for n factorial

Source: https://cse.buffalo.edu/

Permutation as 1-1 Correspondence

A set V consists of n elements if its elements can be counted 1, 2,..., n. In other words, the set V can be brought into a 1-1 correspondence with the set {1, 2, ..., n}. Often, it is more convenient to start counting from 0. Then we get the set {0, 1, 2, ..., n-1}.

A permutation is a 1-1 correspondence of a set V onto itself: f: V → V.

Being able to count elements in the set V means the set can be written as $\{v_1, v_2, ..., v_n\}$. However, a set may be counted in many different ways. For example, a set of two elements can be counted in exactly two ways. The first element first and the second or the first element second and the second first.

In how many ways may one count a set of n elements? Or, which is the same, how many permutations are there of (a set of) n elements?

The number of permutations of a set of n elements is denoted n! (pronounced n factorial)

Thus n! is the number of ways to count a set of n elements. As we saw, 2! = 2. Obviously, 1! = 1, 3! = 6. Indeed, there are just six ways to count three elements:

1. 1, 2, 3
2. 1, 3, 2
3. 2, 1, 3
4. 2, 3, 1
5. 3, 1, 2
6. 3, 2, 1

How many ways are there to count an empty set, the set with 0 elements? (Note that {0} contains one element thus is not empty. The empty set contains no elements at all - {}.) Since there is nothing to count the question is *In how many ways can one do nothing?* A mathematical answer to this is just one: 0! = 1.

Here is another way to do this. Look at the six permutations of a 3-element set. Let us try doing this for a set of n elements. There are n ways to select the first element. For each of these, by definition, the remaining (n-1) elements can be counted in (n-1)! ways. Therefore, there are n·(n - 1)! ways to count an n-element set (Bogolmony, 2018).

Lesson 2. Combination

A **combination** refers to choosing a group of items or selection of objects (Jain, 2017).

- A *r*-combination of elements of a set is an unordered selection of *r* elements from the set.
- So a *r*-combination is simply a subset with r elements.
- The number of *r*-combinations of a set with n elements is denoted by $C(n,r)$ or $\binom{n}{r}$

Example 1:

A very simple example of combinations would be in the number of pizzas one could create given a certain number of criteria. In other words, given that a pizza place has:

- 3 types of crust -thin, regular, thick
- 4 types of toppings - pepperoni, sausage, ham, bacon
- 3 types of cheese - American, cheddar, Swiss

How many different one topping, one cheese pizzas could be made?

Answer - multiply them all together because the order of choice does not matter here.
So, **3*4*3 =36**

Example 2:
How many different ways to select a committee of three members from 10 students?

Solution: Each committee is just a subset of three elements. So, the answer is simply

$C(n,r) = C(10, 3)$

Theorem 2:
For any non-negative integer n and r such that $0 \leq r \leq n$:

$$C(n,r) = \frac{n!}{r!(n-r)!} = \frac{n \cdot (n-1) \cdots (n-r+1)}{r \cdot (r-1) \cdots 2 \cdot 1}$$

Figure 7.2 Formula for any non-negative integer

Source: https://cse.buffalo.edu/

Proof: Consider any r-combination, which is just a r element set A. There are $r!$ different orderings of the elements in A. Each of which is an r -permutation of the set. Thus:

$$P(n,r) = C(n,r) \times r!$$

Since $P(n,r) = \frac{n!}{(n-r)!}$, we have:

$$C(n,r) = \frac{P(n,r)}{r!} = \frac{n!}{r!(n-r)!} = \frac{\overbrace{n \cdot (n-1) \cdots (n-r+1)}^{r \text{ terms}}}{r!}$$

Example 3:
How many different ways to select a committee of three members from 10 students?

Solution: $C(10, 3) = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$

Lesson 3. Real-Life Use of Permutation and Combination

In communication networks, cryptography and network security permutations are frequently used in communication networks and parallel and distributed systems. Routing different permutations on a network for performance evaluation is a common problem in these fields. Many communication networks require secure transfer of information, which drives development in cryptography and network security. This area has recently become particularly significant because of the increased use of internet information transfers. Associated problems include protecting the privacy of transactions and other confidential data transfers and preserving the network security from attacks by viruses and hackers (Rosen, 2018).

Encryption process involves manipulations of sequences of codes such as digits, characters, and words. Hence, they are closely related to combinatorics, possibly with intelligent encryption process. For example, one common type of encryption process is interchanging--i.e., permuting parts of a sequence. Permutations of fast Fourier transforms are employed in speech encryption (Rosen, 2018).

Computer Architecture

Design of computer chips involves consideration of possible permutations of input to output pins. Field-programmable interconnection chips provide user programmable interconnection for a desired permutation. Arrangement of logic gates is a basic element for computer architecture design (Rosen, 2018).

Computational Molecular Biology

This field involves many types of combinatorial and sequencing problems such as atoms, molecules, DNAs, genes, and proteins One-dimensional sequencing problems are essentially permutation problems under certain constraints (Rosen, 2018).

Languages

Both natural and computer languages are closely related to combinatorics. This is because the components of these languages, such as sentences, paragraphs, programs, and blocks, are arrangements of smaller elements, such as words, characters, and atoms. For example, a string searching algorithm may rely on combinatorics of words and characters. Direct applications of this can include word processing and databases (Rosen, 2018).

Another important application area is performance analysis of these string searching algorithms. The study of computability, what we can compute and how it is accomplished draws heavily on combinatorics (Rosen, 2018).

Pattern Analysis

In a broad sense, all the above-mentioned areas can be viewed as special cases of pattern analysis. Molecular biology, for example, studies patterns of atoms, molecules, and DNAs whereas languages treat patterns of sentences, words, and strings. Patterns can have many other forms; for example, visual images, acoustic signals, and other physical quantities such as electrical, pressure, temperature, etc., that appear in engineering problems. Patterns can also be abstract without any associated physical meaning. These patterns may be represented in various ways such as digital, analog, and other units. Some of these types of patterns can be associated with combinatorics. There has been extensive research on combinatorial pattern matching. Computer music can be a specialized application domain of combinatorics of acoustic signals (Combinatorial, 2018).

Scientific Discovery

For certain types of knowledge discovery problems, generation of combinatorial sequences may become necessary in the process of yielding candidate solutions. For example, in scientific discovery, we may want to have a sequence of plausible chemical/biological reactions and their formations. In each step of the sequence, we may generate combinatorial sequences of chemical/biological radicals, bases, and molecular compounds as candidate solutions and may select the most likely ones under certain rules and constraints. In another example, certain areas of mathematics, such as graph theory and number theory, may generate combinatorial sequences as candidate solutions (Rosen, 2018).

Databases and Data Mining

Queries in databases are multiple join operations that are permutations of the constituent join operations. Determining an optimal permutation that gives minimum cost is a common and important problem. Data mining or knowledge discovery in databases is a relatively new field that aims at distilling useful information, often from large databases. In this area, techniques employing symbolic AI can manipulate combinatorial sequences of atoms or information elements (Rosen, 2018).

Non-symbolic knowledge discovery techniques, such as genetic algorithms and genetic programming, most commonly deal with solutions in the form of sequences of bits, digits, characters, and sometimes Lisp program elements. Neural networks, another domain of non-symbolic AI, sometimes deal with combinatorial patterns. Knowledge discovery techniques under uncertainty, such as Bayesian networks, Dempster-Shafer theory, fuzzy logic, and rough set theory, may have combinatorial solutions (Rosen, 2018)

Operations Research

Many optimization problems in operations research (OR) involve combinatorics. The job scheduling problem is essentially a sequencing problem to determine the order of jobs to be processed in an effort to minimize the total time, cost, etc. Here, jobs can be in a computer system, network, or processing plant. Many problems involving graphs or networks also deal with the order of vertices and edges (Rosen, 2018).

The traveling salesperson problem is to determine the order of cities to be visited to minimize the total distance. The shortest path problem of a graph is to determine a sequence of edges, the total length of which is minimum. Oftentimes, these problems are computationally difficult--e.g., NP-complete or NP-hard--and, therefore, require extensive research (Rosen, 2019).

Simulation

Permutations and combinations can be employed for simulations in many areas. Permutations representing various genotype-phenotype associations are employed in genetics simulations (Doerge and Churchill, 1996). Other areas that employ permutations and combinations for simulations include networks, cryptography, databases and OR (Rosen, 2019).

Security

This is a very specialized problem domain that has become a major national challenge after 9/11. To confront this challenge, many intelligent computing techniques have been applied, including intelligent pattern analyses of human faces, X-ray images, chemical components, data from a distributed network of wireless sensors, etc. Natural language processing and data mining techniques have been applied to sift through and monitor the tremendous accumulation of electronic communication data (Rosen, 2019).

Since combinatorics are extensively applied to these intelligent computing techniques, there is a wide spectrum of potentials for the national security issue. Some specific examples may include string searching algorithms and their performance analysis in communication data, pre- and post-analysis of combinatorial sequences of information elements, and combinatorial pattern matching (Rosen, 2019).

Assessment Task

Direction: Kindly answer all the assessment task and submit it on/or before the midterm examination through your account in Edmodo. If hand written, write it legibly and clearly, take a picture of it and send it in a MS Word format. Use the sample filename – CS2108AT_JABanocnoc for BSCS Students and IT2109AT_JABanocnoc for BSIT students:

1. How many poker hands of 5 cards can be selected from a standard 52 cards?
2. Guess the next number in the following sequence
3. There are 5 different books, how many ways can you arrange them on the shelf (in the typical upright position of libraries)?
4. You are a judge in a cat beauty contest. There are 50 cats, and you must pick the prettiest, the 2nd prettiest, and the 3rd prettiest. In how many ways can you do this?
5. You want to buy 3 cats from a pet store that has 50 cats. In how many ways can you do this?
6. Arranging 4 persons in 5 chairs using permutation. Suppose there is a party in your and you have invited 10 close friends in the party as you know that there is only 10 chairs in your house but the real problem occurs when you came to know that you gave 1 chair to your neighbor because some guests have visited on their home and they are short of 1 chair. So, how will you arrange 10 persons in 9 chairs by using the concept of permutation
7. Select 11 players from 15 players team for the NBA Finals 2020. Suppose your team has performed in the NBA Finals 2020 and they are able to qualify for the finals. But a problem occurs, the best shooter has been injured and due to which he will not be able to play for 2 weeks until the finals. So, the team selectors have a choice between Lebron James, Kobe Bryant and Michael Jordan and they have 1 vacant position.
8. Suppose you want to select 5 employees out of 50 employees of your company for the upcoming project, how many ways can you select an employee?

Summary

A permutation is a way of counting elements in a set. What was $\{v_1, v_2, ..., v_n\}$ for one counting is $\{vi_1, vi_2, ..., vi_n\}$ for another. In other words, a permutation is a way of reindexing a set. Most often for the sake of convenience, when doing permutations, indices is all that is considered and the symbol v for the set's element is omitted. This makes sense too. For then we just discussed about permutations of the (index) set {1, 2, 3, ..., n} (Bogolmony, 2018).

So, this is just a basic real-life application of permutation and combination likewise there can be infinite applications like selecting 5 fruits out of 10 fruits, selecting 3 best books our of 50 books for the same subject. The theory probably isn't used so much, but from a practical perspective, when someone is hungry and have to prepare themselves a meal, from all the ingredients in their home, there are many combinations and permutations to consider.

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– END OF MODULE FOR MIDTERM TERM PERIOD –

SUBMIISION OF ASSESSMENT TASKS AND EXAMINATION FOR MIDTERM PERIOD WILL BE ANNOUNCED VIA iLU LMS or GC

MAKE SURE TO CHECK AND DO NOT FORGET TO TAKE IT AS SCHEDULED.